TW

TrackWise

Feature & Demo Manual

Integrated AICTE Approval Workflow Automation,
Compliance Monitoring and Alert Management System

Smart India Hackathon - Round 2 - Smart Automation

Version 1.2 | 27 October 2026

A team guide: what the product does, how it works, how to run it and how to present it.

All institutions, people, thresholds and figures in the product and in this manual are invented for the demo. They are not real AICTE data or norms.

Repository: github.com/AlexDsouza107/trackwise (public)

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What changed in v1.2

This version adds the pieces the problem statement names explicitly: an **Expert Visit Committee (EVC)** stage, **configurable stage routing** per application type, **regions** with regional officers, **lease and prescribed-document validity** tracking with renewal verification, an **admin settings** screen, generated **deficiency letters and case summaries**, **mock integration adapters**, and a **CI pipeline that runs the full Docker stack**. AI scoring stays, but as a bonus signal rather than the headline.

How to use this manual

New to the project? Read sections 1-3, then skim 4. Presenting? Learn section 7 and rehearse it twice. Working on code? Sections 5, 6 and 9 tell you where things live. Everything here was checked against the code in the repository, not written from memory.

1. The 60-second pitch

The problem

Approving an institution, a new course or an intake change is a long chain of hand-offs: documents are checked by hand, expert visits are arranged ad hoc, compliance is looked at separately from the application, institutions often cannot see where their file is, and nobody is warned when a deadline slips or a lease or approval is about to lapse. Problems are found late, and the same file bounces back for missing paperwork.

What TrackWise does about it (in the order a reviewer would ask)

Pillar	What it gives you	Where to see it
Workflow automation	One state machine drives every application. Each application type has its own **configurable route** (only scrutiny, scrutiny + EVC, ...). Illegal moves are impossible and the next owner is notified automatically.	Application detail, Settings
Pendency monitoring	Queues ordered by SLA urgency, overdue highlighting, **region filters** and region-wise performance (open, breached, turnaround, compliance).	Queue, Analytics
Expert Visit Committee	EVC is a real stage: schedule the visit, assign members, file the report, complete or raise a deficiency. Reminders and escalation are automatic.	Expert visits, application detail
Compliance monitoring	Faculty ratio, infrastructure, accreditation and fees scored per institution, trended, and enforced as a **gate**.	Compliance
Lease and document validity	A register of lease and prescribed documents with expiry alerts (90 / 30 / 7 days and on expiry), renewal submission and **renewal verification**.	Document validity
Alerts and escalation	Seven kinds of alert, multi-level escalation with e-mail at higher levels, deduplicated, rules editable.	Alerts
Audit trail and communications	Every action is recorded; **deficiency letters** and **case summaries** are generated from that record.	Audit trail, Communications
AI risk score (bonus)	An explainable document check and risk score. It informs reviewers and never blocks the workflow.	AI assessment card

Say it in one breath

> *"TrackWise turns AICTE approvals from a paper chase into a tracked, automatically escalated pipeline: each application follows the route its type requires, expert visits and document expiries are scheduled and chased by the system, officers always see what is most urgent in their region, and every decision leaves a complete audit trail. An explainable AI score is a bonus on top."*

Three things judges should remember

- **The rules are in the engine, not the UI.** A critical compliance failure literally blocks "Compliance cleared", and the EVC stage cannot be completed without a filed report.
- **Nothing happens silently.** Unscheduled visits, overdue reports, expiring leases and SLA breaches all raise alerts exactly once and escalate up a ladder.
- **It is configurable, not hardcoded.** Routes, stage time limits and compliance thresholds are edited in Settings and take effect without a deployment.

> **Be upfront about what this is**
>
> This is a working demo MVP with illustrative data. Norms, SLAs, regions and documents are invented. The AI is rules-based on purpose and reads metadata, not contents. DigiLocker and the accreditation registry are **mock adapters**. E-mail is a stub. Section 9 lists every limit; stating them yourself earns trust.

2. Who uses it

There are five roles. The server enforces what each may do; the UI renders the actions the server says are available. Regional officers are additionally **scoped to one region**: every list, dashboard and detail page is filtered to it, and other regions return 403. All demo accounts use the password **demo1234**.

Role	Demo accounts	What they do
Institution	inst01 ... inst14 @trackwise.demo (inst13, inst14 are *proposed* institutions)	Create applications, upload or fetch documents (DigiLocker mock), submit, respond to deficiencies, track status, keep the lease and prescribed-document register up to date, renew expiring documents.
Scrutiny officer	officer1, officer2 @trackwise.demo	Work the Document Scrutiny and Compliance Check queues. Verify documents, review AI and compliance results, clear, raise a deficiency (with a drafted reason) or reject. Verify document renewals.
Regional officer	regional1 Northern, regional2 Southern, regional3 Eastern, regional4 Western, regional5 Central	Own the EVC stage in their region: schedule visits, assign committee members, follow up reports, complete the stage. See only their region. Verify renewals in their region.
Committee member	committee1, committee2 @trackwise.demo	Take part in assigned expert visits and file the visit report, then review applications at the committee stage: recommend approval or rejection (note required).
Admin	admin@trackwise.demo	Takes the final decision, sees every region, edits routes, SLA hours, compliance thresholds and escalation rules, runs alert scans, reads the e-mail outbox and analytics.

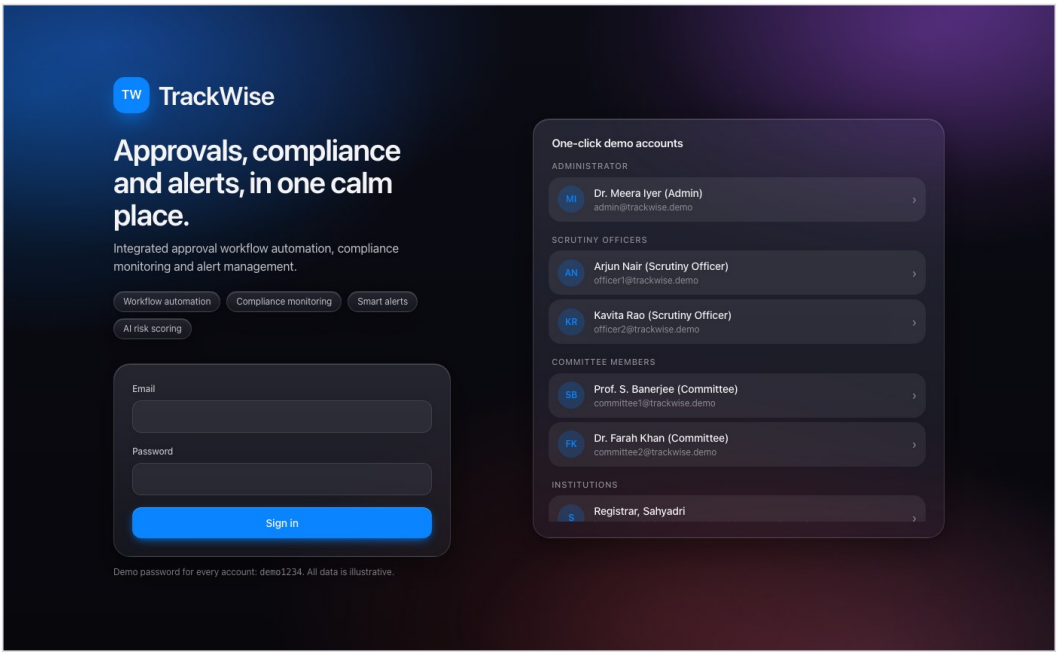
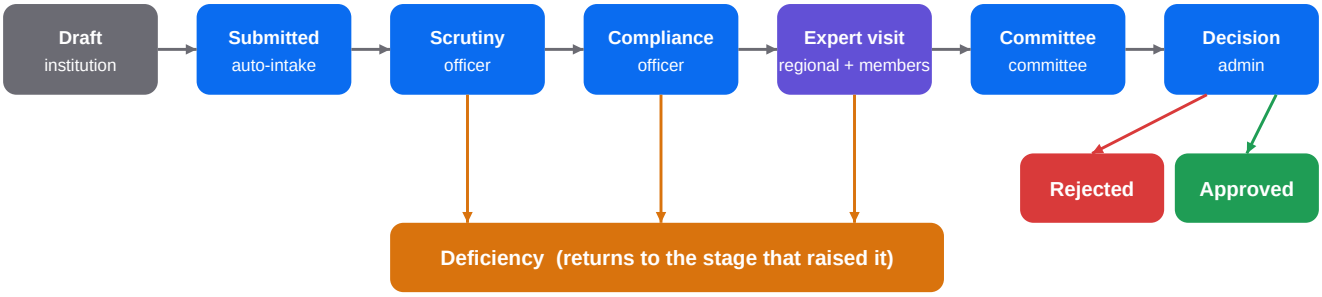


Figure 1 - Login page with one-click demo accounts (dark theme). Each tile signs in instantly.

Tip for presenting: open each role in its own browser profile or private window so you can switch roles without signing out and in.

3. The application lifecycle

Every application is a state machine (XState). The machine says which moves are legal; a separate permission table says which role may trigger each move. What is new is that the **forward path comes from a route** stored per application type, fixed when the application is submitted.



The stages an application passes through depend on its type (configurable route); Decision is always last.

Default routes (illustrative; editable by an admin)

Application type	Route	Why
New institution	Scrutiny > Compliance > **EVC** > Committee > Decision	A physical site and facilities must be inspected.
New course	Scrutiny > Compliance > Committee > Decision	Desk review by the committee is enough.
Intake increase	Scrutiny > Compliance > **EVC** > Committee > Decision	Capacity claims are verified on site.
Extension of approval	Scrutiny > Compliance > Decision	A fast track: no committee, no visit.

Any combination of the four review stages is allowed (for example "only scrutiny" or "only EVC"). A stage outside the route cannot be entered or acted on; the machine tests prove it.

Stages and their SLAs

Stage	Who owns it	SLA (editable)	What happens here
Draft	Institution	-	Fill the form, upload or fetch the required documents (one per type).
Submitted	System	24 h	Automatic intake: the AI document check runs, the route is fixed, the file moves on.
Document Scrutiny	Scrutiny officer	72 h	Check documents and findings. Verify, or raise a deficiency.
Compliance Check	Scrutiny officer	72 h	Rules are evaluated. Critical failures block clearance.
Expert Visit (EVC)	Regional officer, assigned members	336 h	Schedule the visit, assign members, file the report, complete.
Expert Committee Review	Committee	120 h	Recommend approval or rejection, or raise a deficiency.
Decision	Admin	48 h	Final approve or reject. Approval sets a validity date.
Deficiency	Institution	240 h	Fix and resubmit. The file returns to the stage that raised it.
Approved / Rejected	-	-	Final. Audit trail is complete.

Who can do what

Action (event)	Roles allowed	Note?	Notes
Submit application (SUBMIT)	Institution	No	Fixes the route from the current settings.
Documents verified (PASS_SCRUTINY)	Officer, admin	No	Moves to the next stage on the route.
Compliance cleared (CLEAR_COMPLIANCE)	Officer, admin	No	**Guard:** blocked while any critical rule fails.
EVC visit complete (COMPLETE_EVC)	Assigned member, regional officer, admin	Yes	**Guard:** the visit report must be filed.
Recommend approval / rejection (RECOMMEND)	Committee, admin	Yes	Two variants carry the outcome.
Approve / Reject (APPROVE, REJECT)	Admin (reject also officer)	Reject: yes	Approval sets validity.
Raise deficiency (RAISE_DEFICIENCY)	Officer, committee, regional, admin	Yes	Remembers the stage to return to.
Resubmit with corrections (RESUBMIT)	Institution	Yes	Returns to the stage that raised it (EVC starts a new round).

The deficiency loop in one sentence

A reviewer raises a deficiency (the reason can be drafted from the findings); the institution sees a banner and the missing items, fixes them and resubmits with a note; the file goes back to the exact stage that raised it, and the round counter and audit trail record it.

Two gates enforced by the machine, not the screen

Critical compliance failures block "Compliance cleared". The EVC stage cannot be completed until the visit report is on file. Hiding a button would not be enough; the server refuses the event.

4. Feature tour

Sign in and welcome

One-click demo tiles or e-mail and password. After signing in a short welcome sequence plays before the dashboard slides in (skipped when the operating system asks for reduced motion).

Institution portal

- **New application:** choose New Institution, New Course, Intake Increase or Extension of Approval; the form adapts to the type.
- **Documents:** one file per required type. **View**, **+ Add**, the **x** to remove (draft or deficiency only), and **DigiLocker** to fetch an issued document by reference (mock adapter; shown with a "DigiLocker (mock)" badge).
- **Track status:** the stepper shows *this application's own route*, the SLA chip shows time left, and the audit trail lists every event.
- **Document validity:** the application page shows the institution's lease and prescribed-document status, so an expired lease is visible during scrutiny.

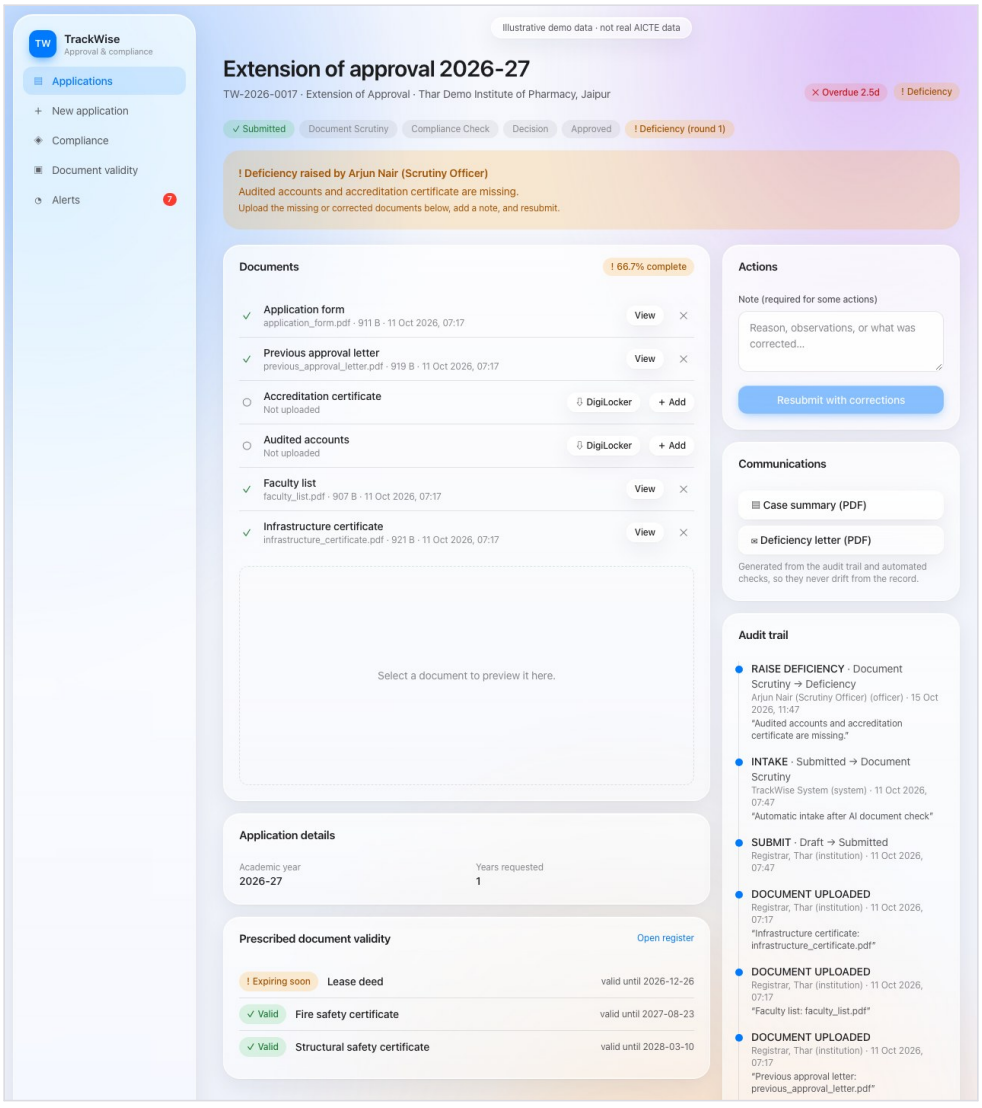


Figure 2 - Institution view of an application in Deficiency: reason banner, checklist with DigiLocker / Add / View / remove, resubmit, communications and document validity.

Queue, regions and pendency

Queues show only the stages a role owns, ordered by SLA urgency with breached items first. Staff can filter by **region**; regional officers are pinned to theirs. The Analytics page adds a **region-wise performance table** (institutions, open, approved, rejected, SLA breached, average turnaround, average compliance).

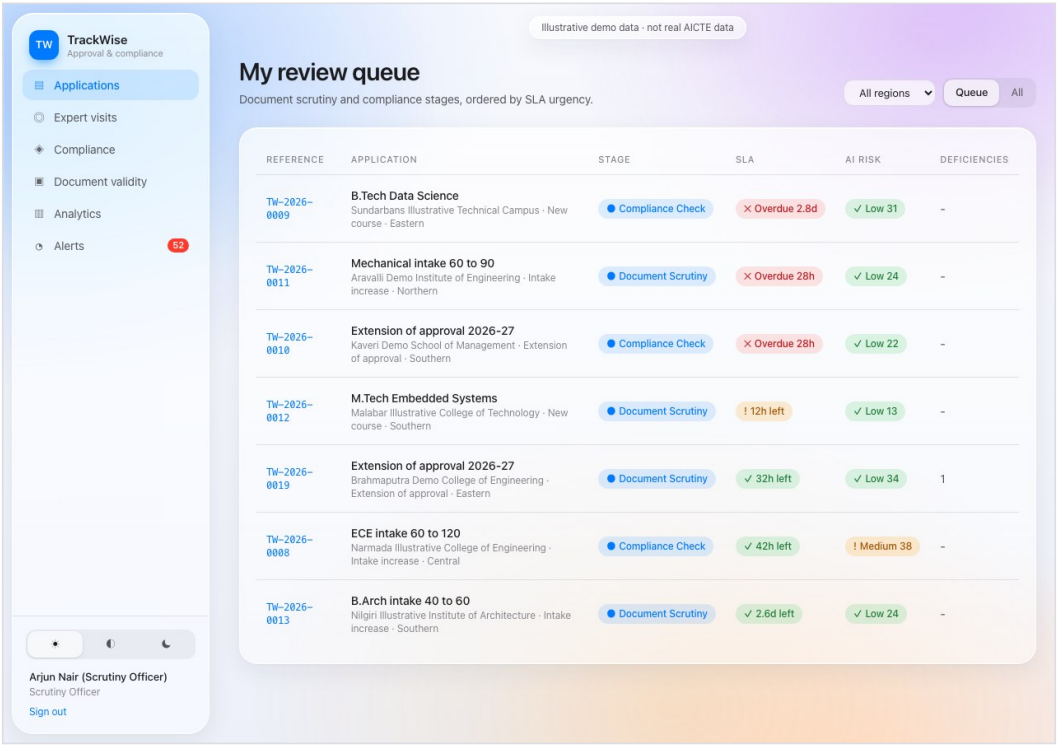


Figure 3 - Scrutiny officer queue. Overdue items pulse and sort first; each row shows the region and the AI risk chip.

Expert visits (EVC)

The **Expert visits** page lists every visit with its status: *needs scheduling*, *in 20h*, *report overdue* or *report filed*. On the application, the regional officer sees a scheduling form (date, venue, committee members); assigned members see the report form.

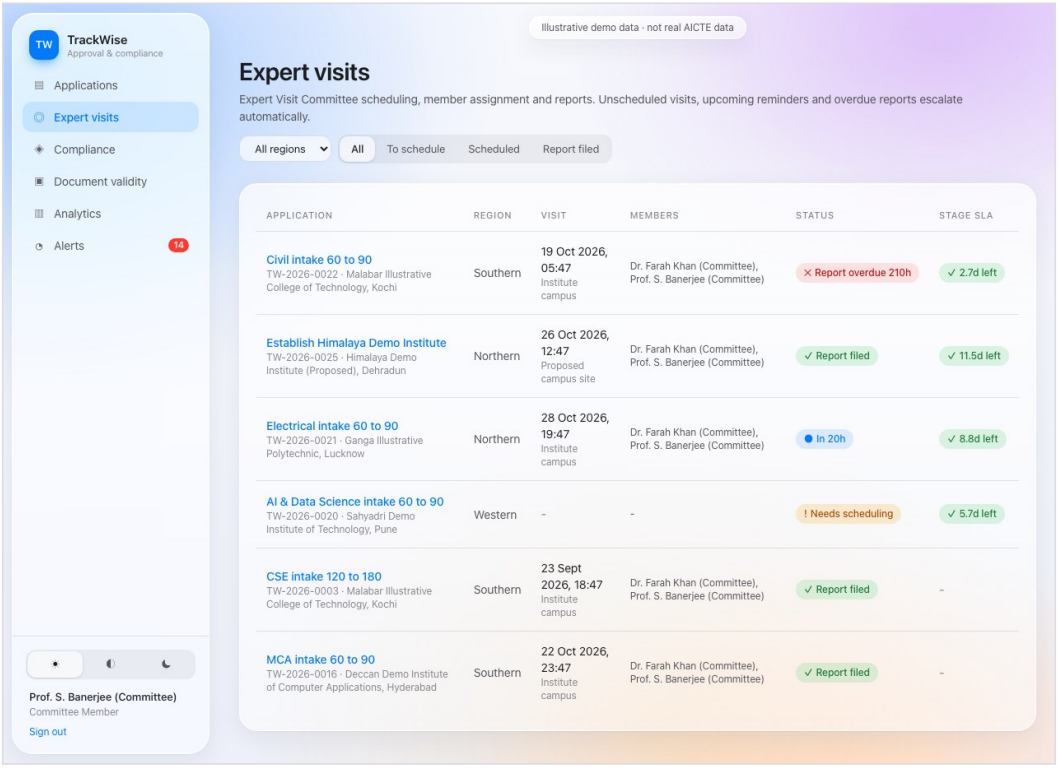


Figure 4 - Expert visits queue across regions: one needs scheduling, one is due in 20 hours, one is 210 hours overdue for its report.

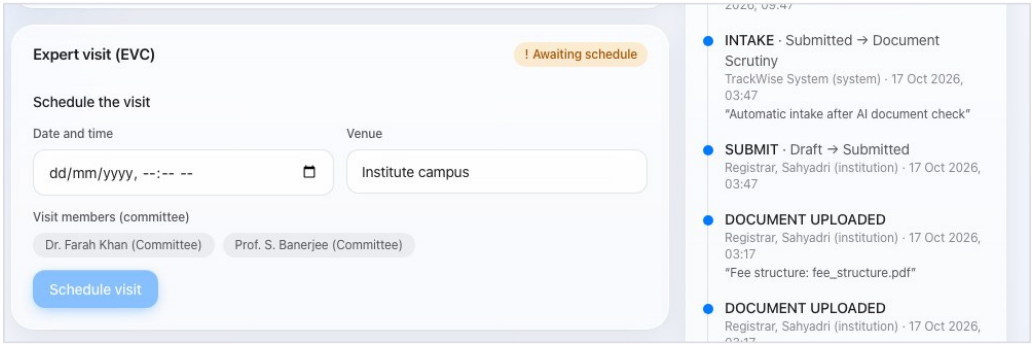


Figure 5 - The regional officer schedules the visit and assigns members from the application page.

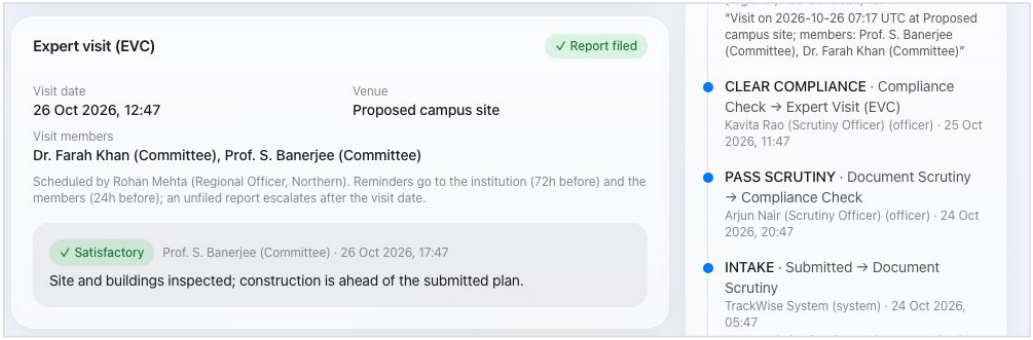


Figure 6 - After the visit: the report (outcome and findings) is on file, and the stage can be completed.

Compliance monitoring

A heatmap shows every institution against every rule with a percentage per cell, the overall score and the change since last month, filterable by region. Selecting an institution shows its six-month trend, the latest rule results, the raw numbers and the **accreditation registry cross-check** (a match, or a mismatch with the difference in days).

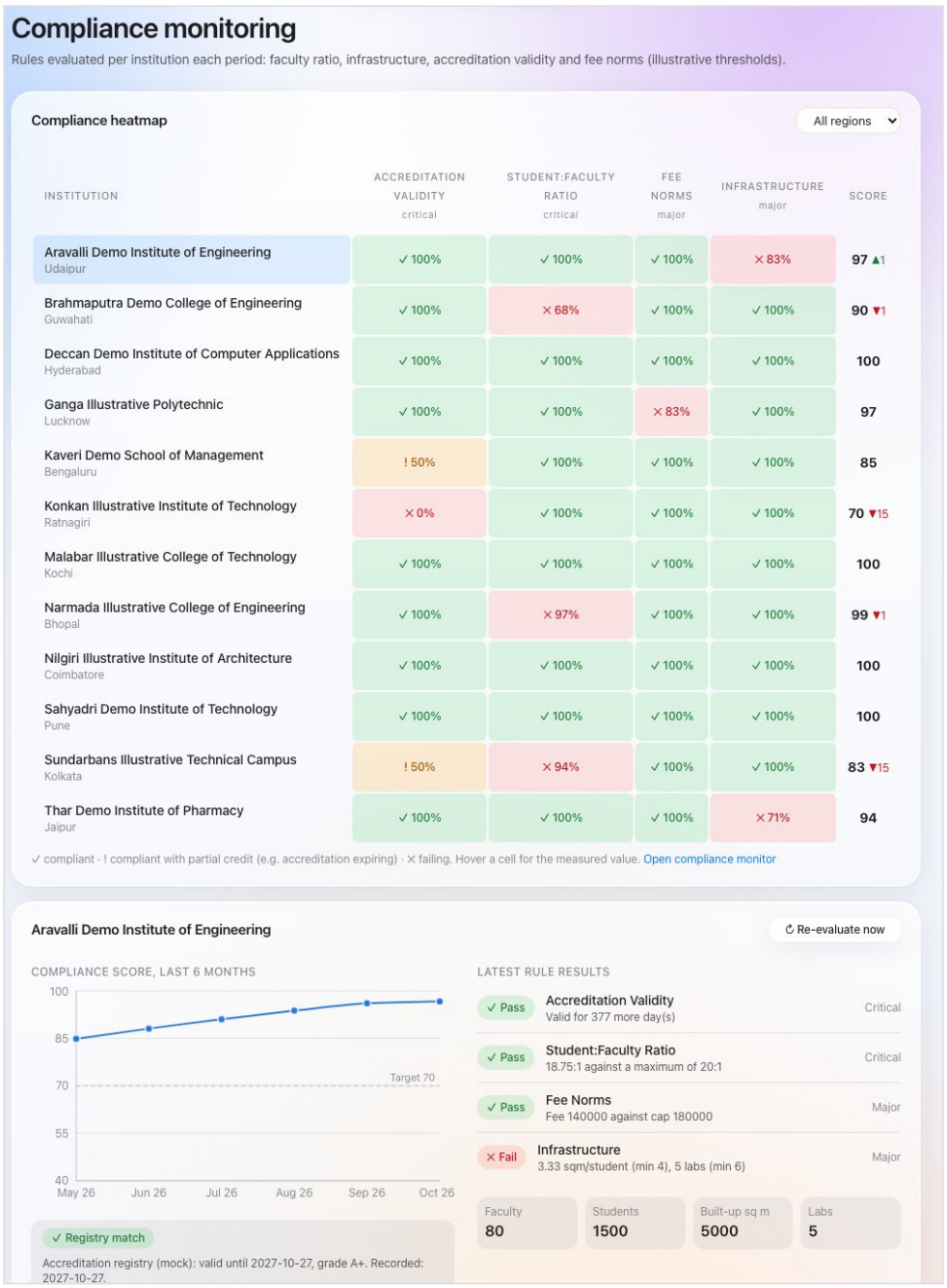


Figure 7 - Compliance heatmap, drill-down and the registry cross-check (mock adapter).

Lease and document validity

Institutions register their lease deed and other prescribed documents with an expiry date. Anything within 90 days is *expiring soon*; past the date it is *expired*. The institution submits a renewal; an officer or regional officer **verifies or rejects** it (a reason is required to reject), and the register keeps the history.

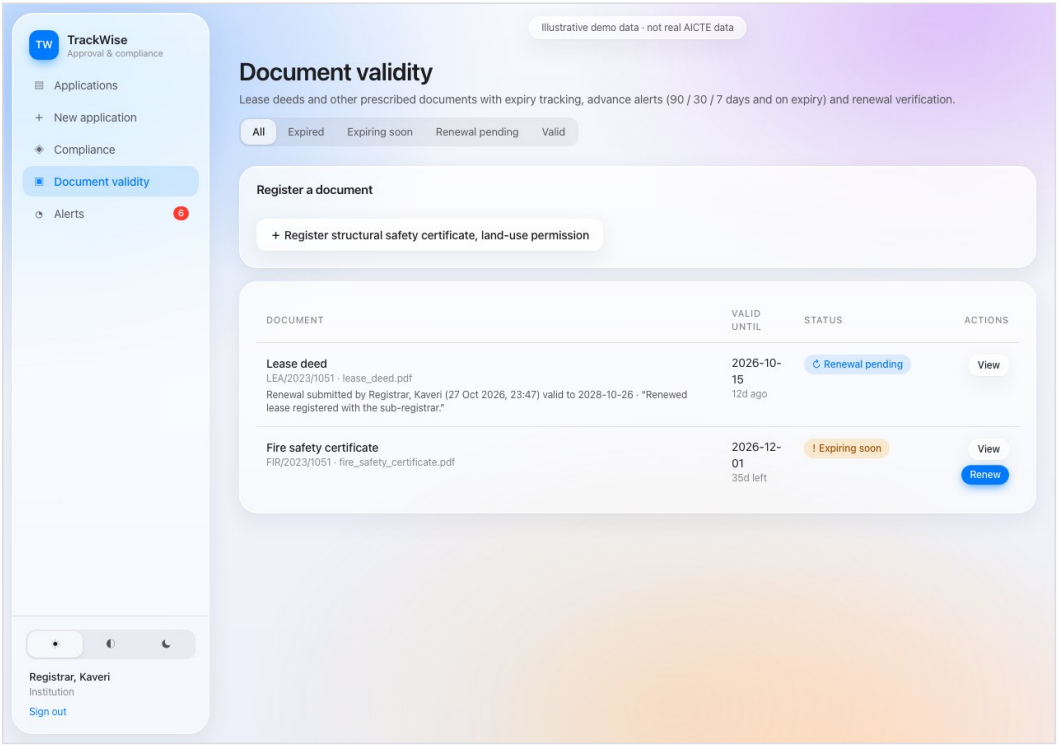


Figure 8 - An institution's register: a lease with a renewal waiting for verification and a fire safety certificate that can be renewed.

Alerts and notifications

My notifications, **Alert log** (each alert with its escalation level), **Escalation rules** (admin edits hours and e-mail per level) and **Email outbox** (stub). **Run alert scan now** triggers the scan on demand for demos; in the full stack a repeatable BullMQ job runs it on a schedule.

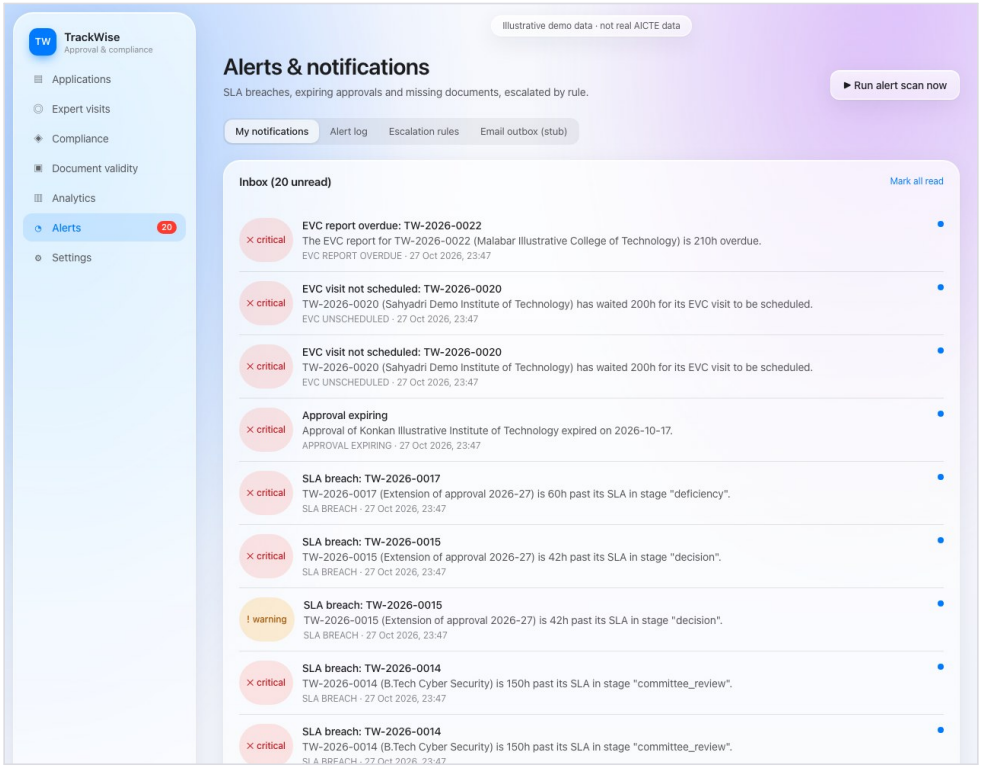


Figure 9 - Admin inbox with escalated alerts.

Analytics

Headline tiles, an EVC and document-expiry strip, the region-wise table, pending applications by stage (including EVC), turnaround by type, AI risk mix, 14-day alert volume and the compliance trend and heatmap.

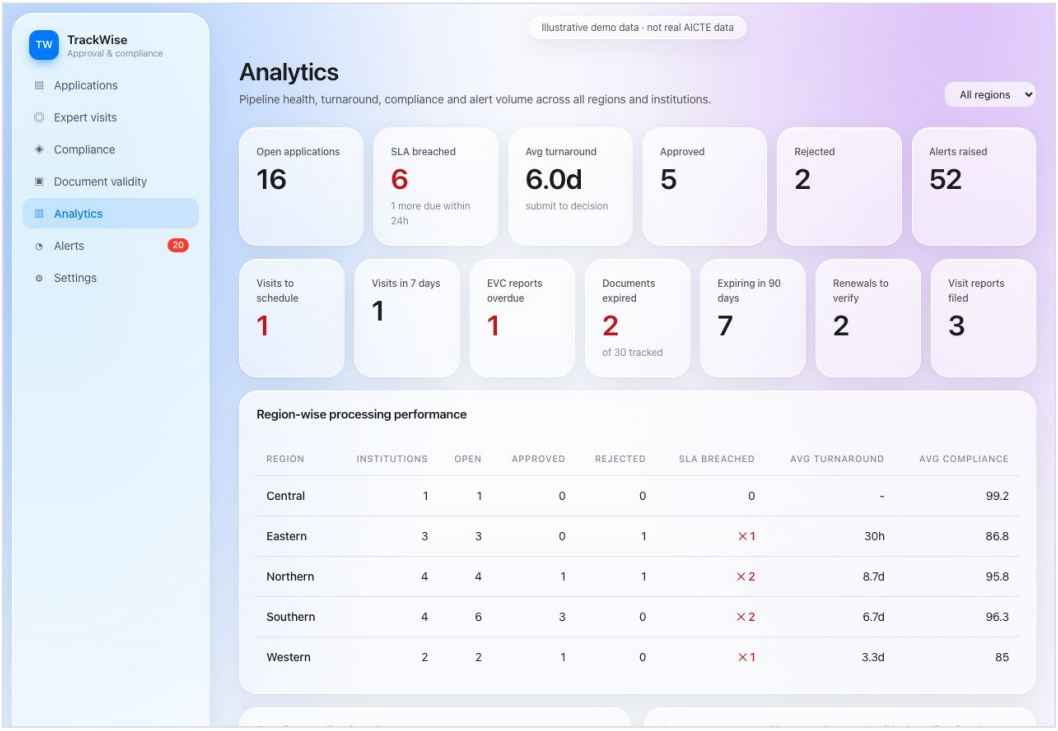


Figure 10 - Analytics: tiles, EVC and document validity counters, and the region-wise performance table.

Settings (admin)

Workflow routes are a checkbox matrix per application type with a live preview of the resulting path. **Stage time limits** are editable per stage. **Compliance rules** have editable thresholds, weights and severity, and a **Re-evaluate all institutions** button. Integrations show which adapters are mocks.

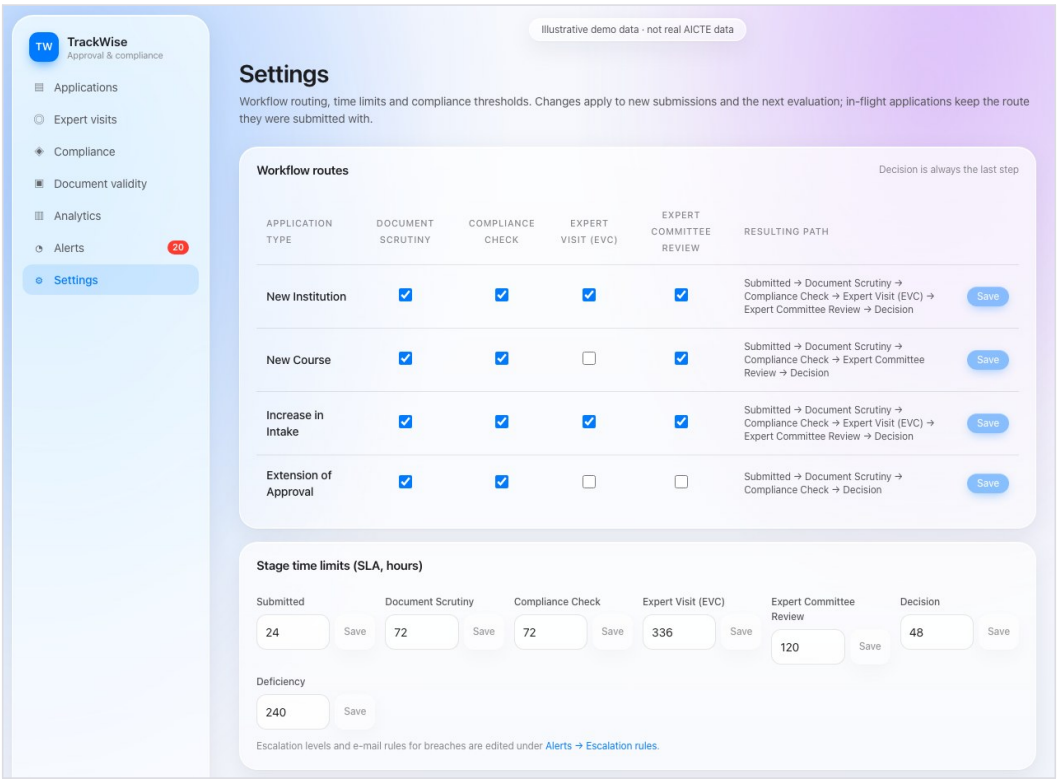
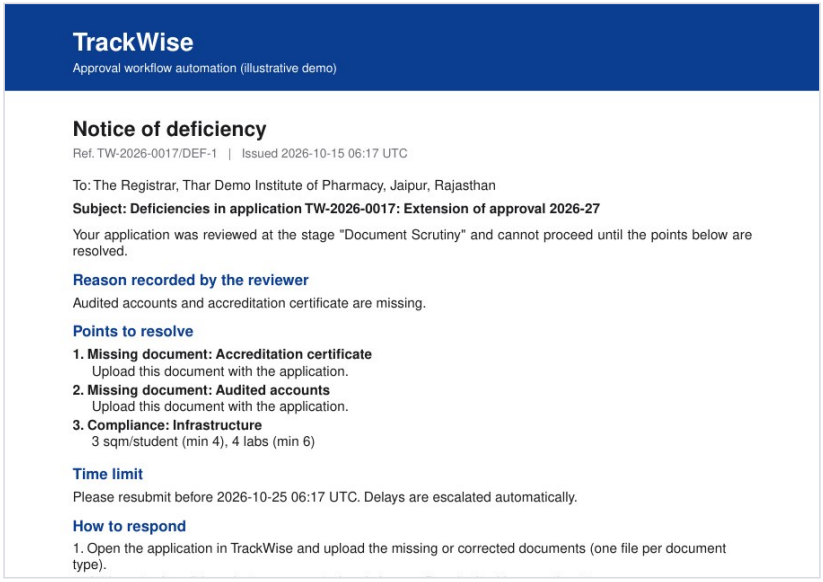


Figure 11 - Settings: workflow routes, SLA hours and compliance rules.

Communications

Draft from findings fills the deficiency reason from the missing documents, failed rules, document findings, EVC report and expired documents. The **deficiency letter** and the **case summary** download as PDFs, generated from the same data, so they never drift from the record.



TrackWise
Approval workflow automation (illustrative demo)

Notice of deficiency
Ref. TW-2026-0017/DEF-1 | Issued 2026-10-15 06:17 UTC

To: The Registrar, Thar Demo Institute of Pharmacy, Jaipur, Rajasthan

Subject: Deficiencies in application TW-2026-0017: Extension of approval 2026-27

Your application was reviewed at the stage "Document Scrutiny" and cannot proceed until the points below are resolved.

Reason recorded by the reviewer

Audited accounts and accreditation certificate are missing.

Points to resolve

1. **Missing document: Accreditation certificate**
   Upload this document with the application.
2. **Missing document: Audited accounts**
   Upload this document with the application.
3. **Compliance: Infrastructure**
   3 sqm/student (min 4), 4 labs (min 6)

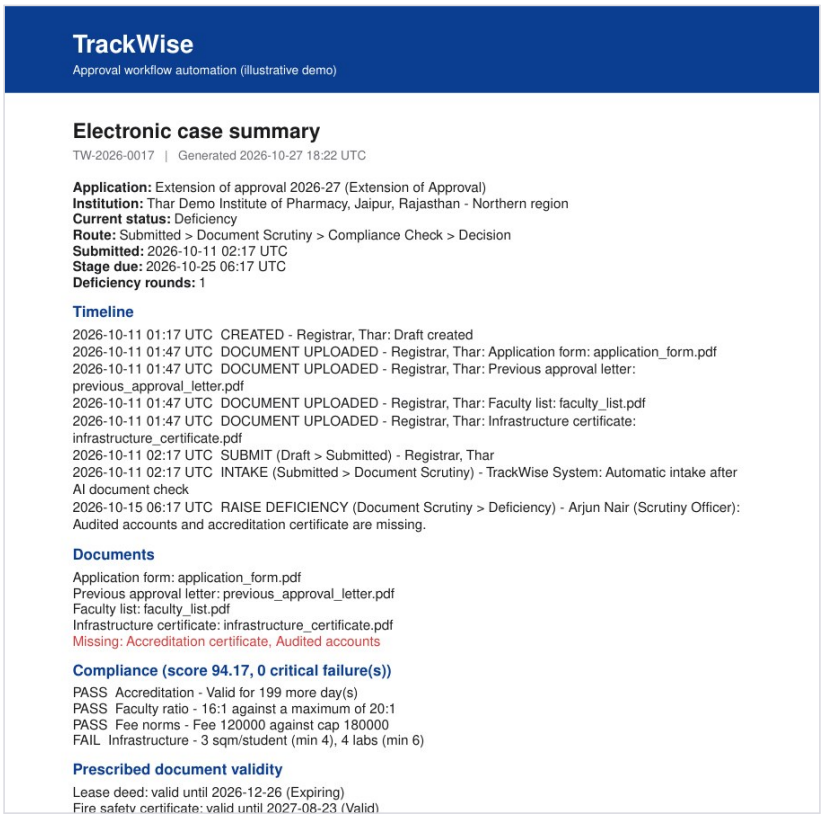
Time limit

Please resubmit before 2026-10-25 06:17 UTC. Delays are escalated automatically.

How to respond

1. Open the application in TrackWise and upload the missing or corrected documents (one file per document type).

Figure 12 - Auto-drafted deficiency letter (PDF).



TrackWise
Approval workflow automation (illustrative demo)

Electronic case summary
TW-2026-0017 | Generated 2026-10-27 18:22 UTC

Application: Extension of approval 2026-27 (Extension of Approval)
Institution: Thar Demo Institute of Pharmacy, Jaipur, Rajasthan - Northern region
Current status: Deficiency
Route: Submitted > Document Scrutiny > Compliance Check > Decision
Submitted: 2026-10-11 02:17 UTC
Stage due: 2026-10-25 06:17 UTC
Deficiency rounds: 1

Timeline

2026-10-11 01:17 UTC CREATED - Registrar, Thar: Draft created
2026-10-11 01:47 UTC DOCUMENT UPLOADED - Registrar, Thar: Application form: application_form.pdf
2026-10-11 01:47 UTC DOCUMENT UPLOADED - Registrar, Thar: Previous approval letter: previous_approval_letter.pdf
2026-10-11 01:47 UTC DOCUMENT UPLOADED - Registrar, Thar: Faculty list: faculty_list.pdf
2026-10-11 01:47 UTC DOCUMENT UPLOADED - Registrar, Thar: Infrastructure certificate: infrastructure_certificate.pdf
2026-10-11 02:17 UTC SUBMIT (Draft > Submitted) - Registrar, Thar
2026-10-11 02:17 UTC INTAKE (Submitted > Document Scrutiny) - TrackWise System: Automatic intake after AI document check
2026-10-15 06:17 UTC RAISE DEFICIENCY (Document Scrutiny > Deficiency) - Arjun Nair (Scrutiny Officer): Audited accounts and accreditation certificate are missing.

Documents

Application form: application_form.pdf
Previous approval letter: previous_approval_letter.pdf
Faculty list: faculty_list.pdf
Infrastructure certificate: infrastructure_certificate.pdf
Missing: Accreditation certificate, Audited accounts

Compliance (score 94.17, 0 critical failure(s))

PASS Accreditation - Valid for 199 more day(s)
PASS Faculty ratio - 16:1 against a maximum of 20:1
PASS Fee norms - Fee 120000 against cap 180000
FAIL Infrastructure - 3 sqm/student (min 4), 4 labs (min 6)

Prescribed document validity

Lease deed: valid until 2026-12-26 (Expiring)
Fire safety certificate: valid until 2027-08-23 (Valid)

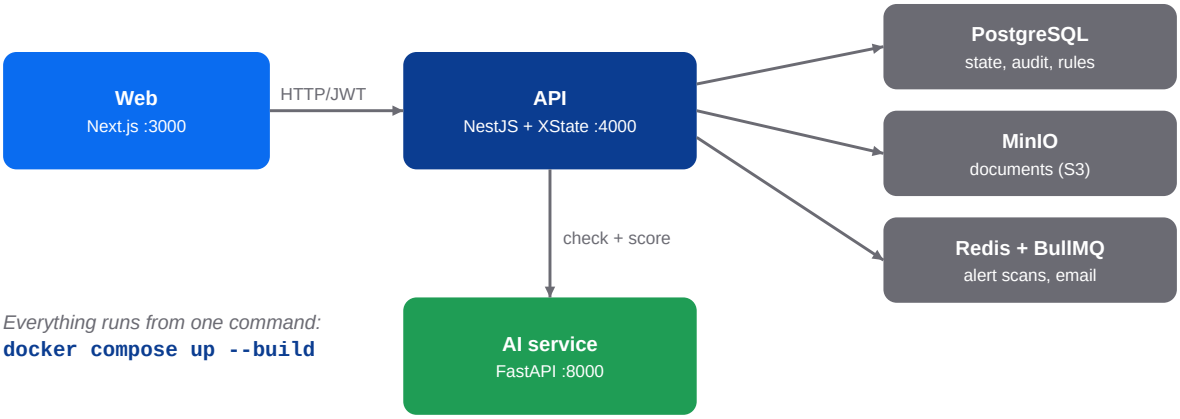
Figure 13 - Electronic case summary (PDF): timeline, documents, compliance, document validity and the AI assessment.

Look and feel

- **Liquid-glass design** in the Apple idiom, **light, auto and dark** themes with a day/night dissolve, and motion that respects "reduce motion".
- **Accessible by design:** status colours are always paired with an icon and a label; the four-series alert chart uses a colour-blind-validated palette with a legend.
- **Demo banner:** "Illustrative demo data - not real AICTE data" is always visible.

5. How it works under the hood

Architecture



Everything runs from one command:
`docker compose up --build`

Layer	Technology	Responsibility
web/	Next.js 14, React 18, Tailwind, Recharts	All screens. Renders the actions the server returns.
api/	NestJS 11, XState 5, zod, pdfkit, JWT	Auth and RBAC, workflow engine, EVC, document validity, compliance, alerts, letters, integrations, configuration, analytics, seeding.
ai/	FastAPI, pydantic v2	/check-documents and /risk-score. Rules-based and explainable.
Data	PostgreSQL 16	Applications, audit log, EVC visits, document register, routes, SLA config, compliance, alerts, notifications.
Files	MinIO (S3 API)	Application documents and register documents; streamed through the API so access control stays in one place.
Jobs	Redis + BullMQ	Repeatable alert scan and e-mail jobs.

Configurable routing

Each application type has a row in `workflow_routes` (an ordered list of review stages). On **Submit**, the current route is written into the application's context, so changing a route later never alters applications already in flight. The machine's forward transitions are generated for every possible next stage, and a guard picks the one the route says comes next. SLA hours live in `sla_config` and are read at each transition.

Expert Visit Committee (EVC)

Step	Who	Rule enforced by the API
Stage entered	System	A visit record is opened (round 1; a re-entry after a deficiency starts the next round). The regional officer of the region is notified.
Schedule	Regional officer (own region) or admin	Date must not be in the past; members must be committee users; the institution and members are notified (e-mail to the institution).
Report	Assigned member, regional officer, admin	Needs a scheduled visit; outcome (satisfactory / minor gaps / major gaps) and findings of at least 10 characters.
Complete	Assigned member, regional officer, admin	The machine guard requires a filed report; the next stage is whatever the route says.
Deficiency	Officer, committee, regional, admin	Returns here after resubmission, as a new visit round.

Document and lease validity

The register has one entry per institution and document kind (lease deed, fire safety, structural safety, land-use permission). Status is computed from the expiry date: *valid*, *expiring* (90 days or fewer), *expired*, or *renewal pending* while a renewal waits for verification. A renewal must extend validity beyond the current date. Approving it replaces the document and keeps the old dates in the history; rejecting it needs a reason and leaves the original untouched. Alerts are suppressed while a renewal is pending, and the verifier is notified instead.

Compliance engine

Each rule returns pass or fail, a score between 0 and 1 (partial credit), the measured value and the threshold. The institution score is the weight-averaged rule score times 100. The four rules, their severity, weight and thresholds are editable in Settings; **Re-evaluate all institutions** applies a change to existing results.

Rule	Severity	Weight	Default threshold (illustrative)
Student:faculty ratio	critical	3	No more than 20 students per faculty member
Infrastructure	major	2	At least 4 sq m per student and 6 labs
Accreditation validity	critical	3	Valid; partial credit under 180 days; 0 when lapsed
Fee norms	major	2	Annual fee not above the cap

Alerts and escalation

One scan covers seven kinds of problem. Each (entity, level) pair raises **once** (deduplicated by key), so scanning twice never spams anyone. Hours mean different things per kind, and an edit of a level in Alerts takes effect on the next scan.

Kind	Level 1	Level 2	Level 3 / 4
SLA breach (hours past due)	0h: stage owner	24h: admin + e-mail	72h: admin + e-mail
Approval expiring (hours before)	90 days: institution	30 days: institution + e-mail	30 days: admin
Document expiring (hours before)	90 days: institution	30 days: institution + e-mail	7 days: regional officer; on expiry: officer + e-mail
Missing documents (hours in stage)	24h: institution	96h: institution + e-mail	168h: officer
EVC not scheduled (hours in stage)	48h: regional officer	96h: admin + e-mail	168h: admin + e-mail
EVC visit reminder (hours before)	72h: institution	24h: visit members + e-mail	-
EVC report overdue (hours after)	24h: visit members	72h: regional officer + e-mail	120h: admin + e-mail

E-mail is a stub: a row in the outbox table and a BullMQ job that logs "[EMAIL STUB]". A rescheduled visit re-arms its reminders.

Communications

The **draft note**, the **deficiency letter** and the **case summary** are built from one place (the application view: audit trail, document check, compliance outcomes, EVC report, document validity) and rendered to PDF with pdfkit. The letter lists the reason, the points to resolve, the time limit and how to respond.

Integrations (mocks)

DigiLocker-style fetch: the institution enters a reference and the adapter returns a clearly labelled demo PDF; the document is stored with its source and the audit trail records "fetched via DigiLocker (mock)". References ending in 000 simulate "not found". **Accreditation registry:** returns the registry's validity for an institution and compares it with the recorded date; Kaveri is a deliberate mismatch. Both sit behind small interfaces in `api/src/integrations`, so a real client replaces the mock without touching the workflow.

AI service (bonus signal)

Document check (metadata only) flags missing documents, unsupported formats, tiny files, duplicate content. **Risk score** is additive: baseline 10 plus itemised factors (type, missing documents, low compliance score, faculty ratio, intake increase, fee, accreditation, prior deficiencies), capped at 100; Low below 35, Medium below 65. If the AI service is down, the workflow continues and the panel says so. Treat it as decision support, not as a verdict.

Design decisions worth knowing

- **The machine decides validity; the API decides permission.** The UI renders `availableActions` from the server, so buttons cannot drift from the rules.
- **Transitions are transactional.** The row is locked, the audit row is written, then entry side effects run (AI refresh, EVC record, notifications, approval validity), and may chain further events.
- **The route is frozen at submit.** Settings changes never alter in-flight applications.
- **Region scoping is in the API** (one helper), not in the UI, so a regional officer cannot widen a filter or open another region's file.
- **Seed data goes through the real engine** with back-dated timestamps, so EVC reminders, overdue reports and escalations are genuine.

6. Running and testing it

Option A - full stack with Docker (the intended way)

```
docker compose up --build
```

Service	URL	Notes
Web app	http://localhost:3000	Sign in with any demo account
API	http://localhost:4000/health	Health and record count
AI service docs	http://localhost:8000/docs	Swagger UI
MinIO console	http://localhost:9001	Demo credentials are in the README

On first boot the API creates the schema and seeds the demo data in a few seconds. Then run `./infra/smoke.sh` to check the whole stack.

> **Honest status of the Docker path**
>
> The compose stack has **not been run on the development machine** (no Docker installed there). Instead: every service was verified natively, `infra/smoke.sh` passes against the no-Docker stack, and the repository carries a GitHub Actions pipeline that builds all images and runs the real six-service stack plus the smoke test on every push. **Before presenting, check that the latest CI run on GitHub is green, or run the command above once on a machine with Docker.**

Option B - without Docker (verified)

You need Node 20+, Python 3.11+ and one PostgreSQL database. MinIO and Redis are replaced by in-memory stand-ins, so uploaded documents disappear on restart and alert scans are triggered with the "Run alert scan now" button.

```
DATABASE_URL=postgres://user@127.0.0.1:5432/trackwise ./infra/dev-lite.sh
```

Resetting the demo data

In no-Docker mode drop and recreate the database and start the API again (it reseeds; an upgrade from v1.1 needs a fresh database because the seed changed). In Docker, `docker compose down -v` removes the volumes.

Tests and CI

Suite	Command	Covers
AI service (3)	cd ai && .venv/bin/python -m pytest	Completeness, findings, risk explanation adds up
API unit (11)	cd api && npm test	Compliance engine; workflow machine including routing, EVC guard and deficiency return
API end-to-end (23)	cd api && npm run test:e2e	Seed, routes, EVC flow, regions, document validity, letters, integrations, configuration, RBAC, alerts, analytics
Smoke (running stack)	./infra/smoke.sh	Health, sign-in, AI reachable, regions and visits, storage round-trip, PDF, web

`.github/workflows/ci.yml` runs the AI tests, API typecheck, unit and end-to-end tests (real Postgres, real AI service), the web build, and then `docker compose up -d --build` followed by the smoke test.

7. The demo script

Total about 12 minutes (A to C are the core eight). Seeded applications are found by title; reference numbers look like TW-2026-0008 and depend on when the data was seeded. Lines in blue are suggested things to say.

Before you start

- Three browser profiles or private windows: institution, officer or committee, admin. Open a fourth for `regional4` if you show EVC.
- Check `/health` returns a number of applications, or run `./infra/smoke.sh`.
- Do one dry run so nothing about the first load surprises you.

A. Happy path (about 3 minutes)

Step	Sign in as	Do this	Say this
1	inst01	New application, New Course. Title "B.Tech Quantum Computing", UG, intake 30. Create draft.	"An institution starts a request in under a minute."
2	inst01	Upload all five documents (any small PDF). Submit.	"On submit the AI checks the documents, the type's route is fixed, and the file moves into scrutiny."
3	officer1	Queue (breached first). Open it, View a document, point at the AI card. Documents verified, then Compliance cleared.	"Officers see the most urgent work first, in their own words."
4	committee1	Open it, Recommend approval with a note.	"The committee stage needs a recorded reason."
5	admin	Approve. Open the audit trail and the Case summary PDF.	"Every step is on the record, and the summary is generated from it."

B. Deficiency loop (about 2 minutes)

Step	Sign in as	Do this	Say this
1	inst05	Open "Extension of approval 2026-27" (Thar). Read the banner, open the Deficiency letter PDF.	"The institution sees exactly what is wrong, in a standard letter."
2	inst05	Click DigiLocker on Audited accounts, enter any reference, Fetch. Add the other document, write a note, Resubmit.	"Documents can be fetched from an authorised source, and the file goes back to the stage that raised it."
3	officer1	The file is back in the queue with Deficiencies = 1. On another application click Raise deficiency, then Draft from findings.	"The reason drafts itself from the findings; the officer edits it."

C. Compliance gate, alerts and analytics (about 3 minutes)

Step	Sign in as	Do this	Say this
1	officer1	Open "ECE intake 60 to 120" (Narmada). Compliance shows a critical ratio failure; only Raise deficiency is offered.	"Critical failures are enforced by the state machine, not by hiding buttons."
2	officer1	Compliance page: region filter, heatmap, Konkan; point at the registry cross-check on Kaveri.	"Compliance is monitored continuously, and checked against the registry."
3	admin	Alerts: log, escalation rules (edit hours), Run alert scan now twice.	"Three levels of escalation; the second scan raises nothing."
4	admin	Analytics: pick a region, show the region-wise table and the EVC and document counters.	"This is the pendency picture per region."

D. Expert visit (EVC) and regions (about 2 minutes)

Step	Sign in as	Do this	Say this
1	regional4 (Western)	Expert visits: "AI & Data Science intake" needs scheduling. Open it, pick a date, venue and members, Schedule.	"Unscheduled visits had already escalated; now reminders are armed."
2	committee1	Expert visits: Himalaya has its report filed. Open it, click EVC visit complete.	"The stage cannot complete without the report."
3	regional1 (Northern)	Only Northern applications are visible; open a Western one by URL: 403.	"Regional officers are scoped to their region by the API."

E. Lease and document validity (about 1 minute)

Step	Sign in as	Do this	Say this
1	inst03 (Kaveri)	Document validity: expired lease with a renewal waiting; the fire certificate is expiring, click Renew.	"The institution keeps its leases alive; the system chases it."
2	officer1	Document validity: Verify the renewal (or Reject with a reason).	"Renewals are verified, with history."

F. Configuration (about 1 minute)

Step	Sign in as	Do this	Say this
1	admin	Settings: untick Compliance Check for Extension of Approval, Save. Change the faculty ratio and Re-evaluate all institutions.	"Routes and thresholds are data, not code."

> **If something goes wrong on stage**
>
> Refresh once. If a document will not open, say that uploaded files live in memory in the no-Docker mode and use a seeded application instead. Keep a screen recording of the full demo as a backup.

8. Questions judges will ask

Is this configurable, or hardcoded?

Routes per application type, stage time limits, compliance thresholds, weights and severity, and escalation rules are all edited in the UI and stored in the database. Changing a route affects only applications submitted afterwards.

How does the EVC work?

It is a stage in the state machine. The regional officer schedules the visit and assigns committee members, they file the report, and only then can the stage complete. Unscheduled visits, upcoming visits and overdue reports each have escalation rules.

What about regions?

Institutions carry a region and regional officers are bound to one. The API filters every list and refuses other regions with 403; analytics adds a region-wise performance table. The region grouping of states is illustrative.

How do you track leases and other documents?

A register per institution with expiry dates, alerts at 90 / 30 / 7 days and on expiry, renewal submission and verification by an officer, and a history. It is separate from compliance scoring on purpose.

Is the AI real?

It is an explainable, rules-based model by design and a bonus signal: officers see every factor behind a score and nothing blocks on it. Document contents are not read yet; OCR / LLM extraction and a trained model are the roadmap.

Which integrations exist?

Two mock adapters (DigiLocker-style fetch, accreditation registry) and an e-mail stub, each behind a small interface. We label them as mocks everywhere; replacing one does not touch the workflow.

How is it secure?

JWT authentication, server-side RBAC and ownership checks on every endpoint, region scoping, validated input (zod, returning 400), and documents served only through the API. Demo limits: shared passwords, tokens in local storage, no rate limiting yet.

Is the data real?

No. Institutions, people, norms, SLAs, regions and documents are invented and marked as illustrative in the app and the README.

Does Docker work?

It is built and smoke-tested by CI on every push (see the latest run on GitHub); on the development machine it was not run because Docker was not installed, but every service and the smoke test were verified natively.

What did you test?

3 AI tests, 11 unit tests and 23 end-to-end tests against a real database and the real AI service, plus a smoke test of a running stack.

9. Limits, roadmap and repo map

What is a stand-in

Area	Today	Production would use
Norms, SLAs, routes, regions	Illustrative placeholders, editable in Settings	The official handbook, loaded as configuration
AI	Rules-based, metadata only; a bonus signal	OCR / LLM extraction plus a trained model with explanations
Integrations	Mock DigiLocker and accreditation registry	Real, authorised AICTE and government APIs behind the same interfaces
E-mail	Outbox table and a logging job	SMTP or an SMS/WhatsApp gateway
Auth	Demo JWT, shared passwords, local storage	SSO, httpOnly cookies, refresh tokens, rate limiting
Regions	State-to-region grouping; officers national, regional officers scoped	The real regional office structure and bureau roles
Docker stack	Built and smoke-tested in CI; not run on the dev machine	Same pipeline, plus deployment manifests

Roadmap ideas, in priority order

- Deploy a public demo and keep the CI badge green.
- Document intelligence: OCR / LLM extraction of faculty count, land area and dates; mismatch flags against the form; a trained risk model on real history.
- Bureau-level roles and dashboards; real regional structure.
- Real e-mail/SMS, a public status tracker, multilingual UI, tamper-evident audit log, load testing.

Repo map

Where	What to look for
api/src/workflow/machine.ts	States, events, route-driven transitions, guards, default routes, the ACTIONS (role) table
api/src/workflow/applications.service.ts	Transition dispatch (transaction and side effects), documents, audit log
api/src/workflow/evc.service.ts	Scheduling, member assignment, visit report
api/src/workflow/config.service.ts	Editable routes and SLA hours
api/src/validity/	Lease and prescribed-document register: renewal and verification
api/src/compliance/engine.ts	The four rules and the score formula
api/src/alerts/scanner.service.ts	Escalation ladder and all seven scans
api/src/letters/	Draft note, deficiency letter and case summary (PDF)
api/src/integrations/	Mock adapters behind interfaces
api/src/seed/seed.service.ts	Illustrative institutions, scenarios and documents, seeded through the real engine
api/test/e2e.spec.ts	End-to-end scenarios
web/app/(app)/...	Screens: applications, evc, documents, compliance, analytics, alerts, settings
infra/smoke.sh, .github/workflows/ci.yml	Smoke test and the CI pipeline (including the Docker stack)

Glossary

Term	Meaning
EVC	Expert Visit Committee: members who visit an institution, then file a report on what they found.
Route	The ordered review stages an application type passes through before the decision.
SLA	The time a stage is allowed to take. Past it, the application is overdue and an alert fires.
Deficiency	A formal request for the institution to fix or add something before the file can continue.
Escalation	Repeating an alert to a more senior role as the delay grows.
Guard	A condition on a state-machine transition; if false, the transition is refused.
Dedupe key	A unique key that stops the same alert being raised twice.